

PRACTICAL NO.:06

Batch:C2

Problem Statement:

Implement K-Means clustering/ hierarchical clustering on sales_data_sample.csv dataset.
Determine the number of clusters using the elbow method.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("sales_data_sample.csv",encoding='latin1')
```

```
df
```

	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER
SALES \				
0	10107	30	95.70	2
2871.00				
1	10121	34	81.35	5
2765.90				
2	10134	41	94.74	2
3884.34				
3	10145	45	83.26	6
3746.70				
4	10159	49	100.00	14
5205.27				
...	...	...	...	...
.				..
2818	10350	20	100.00	15
2244.40				
2819	10373	29	100.00	1
3978.51				
2820	10386	43	100.00	4

5417.57							
2821	10397		34	62.24		1	
2116.16							
2822	10414		47	65.52		9	
3079.44							
	ORDERDATE	STATUS	QTR_ID	MONTH_ID	YEAR_ID	...	\
0	2/24/2003 0:00	Shipped	1	2	2003	...	
1	5/7/2003 0:00	Shipped	2	5	2003	...	
2	7/1/2003 0:00	Shipped	3	7	2003	...	
3	8/25/2003 0:00	Shipped	3	8	2003	...	
4	10/10/2003 0:00	Shipped	4	10	2003	...	
...	...	...	...	...	...	...	
2818	12/2/2004 0:00	Shipped	4	12	2004	...	
2819	1/31/2005 0:00	Shipped	1	1	2005	...	
2820	3/1/2005 0:00	Resolved	1	3	2005	...	
2821	3/28/2005 0:00	Shipped	1	3	2005	...	
2822	5/6/2005 0:00	On Hold	2	5	2005	...	
	ADDRESSLINE1	ADDRESSLINE2				CITY	STATE
\							
0	897 Long Airport Avenue	NaN				NYC	NY
1	59 rue de l'Abbaye	NaN				Reims	NaN
2	27 rue du Colonel Pierre Avia	NaN				Paris	NaN
3	78934 Hillside Dr.	NaN				Pasadena	CA
4	7734 Strong St.	NaN				San Francisco	CA
...	...	...				...	...
2818	C/ Moralzarzal, 86	NaN				Madrid	NaN
2819	Torikatu 38	NaN				Oulu	NaN
2820	C/ Moralzarzal, 86	NaN				Madrid	NaN
2821	1 rue Alsace-Lorraine	NaN				Toulouse	NaN
2822	8616 Spinnaker Dr.	NaN				Boston	MA
	POSTALCODE	COUNTRY	TERRITORY	CONTACTLASTNAME	CONTACTFIRSTNAME		
DEALSIZE							
0	10022	USA	NaN	Yu	Kwai		
Small							
1	51100	France	EMEA	Henriot	Paul		
Small							
2	75508	France	EMEA	Da Cunha	Daniel		

Medium					
3	90003	USA	NaN	Young	Julie
Medium					
4	NaN	USA	NaN	Brown	Julie
Medium					
...	...	...	...	...	...
...					
2818	28034	Spain	EMEA	Freyre	Diego
Small					
2819	90110	Finland	EMEA	Koskitalo	Pirkko
Medium					
2820	28034	Spain	EMEA	Freyre	Diego
Medium					
2821	31000	France	EMEA	Roulet	Annette
Small					
2822	51003	USA	NaN	Yoshido	Juri
Medium					

[2823 rows x 25 columns]

df.head()

	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	
SALES \					
0	10107	30	95.70	2	2871.00
1	10121	34	81.35	5	2765.90
2	10134	41	94.74	2	3884.34
3	10145	45	83.26	6	3746.70
4	10159	49	100.00	14	5205.27

	ORDERDATE	STATUS	QTR_ID	MONTH_ID	YEAR_ID	...	\
0	2/24/2003 0:00	Shipped	1	2	2003	...	
1	5/7/2003 0:00	Shipped	2	5	2003	...	
2	7/1/2003 0:00	Shipped	3	7	2003	...	
3	8/25/2003 0:00	Shipped	3	8	2003	...	
4	10/10/2003 0:00	Shipped	4	10	2003	...	

	ADDRESSLINE1	ADDRESSLINE2	CITY	STATE	\
0	897 Long Airport Avenue	NaN	NYC	NY	
1	59 rue de l'Abbaye	NaN	Reims	NaN	
2	27 rue du Colonel Pierre Avia	NaN	Paris	NaN	
3	78934 Hillside Dr.	NaN	Pasadena	CA	
4	7734 Strong St.	NaN	San Francisco	CA	

POSTALCODE	COUNTRY	TERRITORY	CONTACTLASTNAME	CONTACTFIRSTNAME
DEALSIZE				

0	10022	USA	NaN	Yu	Kwai
Small					
1	51100	France	EMEA	Henriot	Paul
Small					
2	75508	France	EMEA	Da Cunha	Daniel
Medium					
3	90003	USA	NaN	Young	Julie
Medium					
4	NaN	USA	NaN	Brown	Julie
Medium					

[5 rows x 25 columns]

df.tail()

	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER
SALES \				
2818	10350	20	100.00	15
2244.40				
2819	10373	29	100.00	1
3978.51				
2820	10386	43	100.00	4
5417.57				
2821	10397	34	62.24	1
2116.16				
2822	10414	47	65.52	9
3079.44				

	ORDERDATE	STATUS	QTR_ID	MONTH_ID	YEAR_ID	...	\
2818	12/2/2004 0:00	Shipped	4	12	2004	...	
2819	1/31/2005 0:00	Shipped	1	1	2005	...	
2820	3/1/2005 0:00	Resolved	1	3	2005	...	
2821	3/28/2005 0:00	Shipped	1	3	2005	...	
2822	5/6/2005 0:00	On Hold	2	5	2005	...	

	ADDRESSLINE1	ADDRESSLINE2	CITY	STATE	POSTALCODE
COUNTRY \					
2818	C/ Moralarzal, 86	NaN	Madrid	NaN	28034
Spain					
2819	Torikatu 38	NaN	Oulu	NaN	90110
Finland					
2820	C/ Moralarzal, 86	NaN	Madrid	NaN	28034
Spain					
2821	1 rue Alsace-Lorraine	NaN	Toulouse	NaN	31000
France					
2822	8616 Spinnaker Dr.	NaN	Boston	MA	51003
USA					

	TERRITORY	CONTACTLASTNAME	CONTACTFIRSTNAME	DEALSIZE
2818	EMEA	Freyre	Diego	Small

2819	EMEA	Koskitalo	Pirkko	Medium
2820	EMEA	Freyre	Diego	Medium
2821	EMEA	Roulet	Annette	Small
2822	NaN	Yoshido	Juri	Medium

[5 rows x 25 columns]

df.describe()

	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER \
count	2823.000000	2823.000000	2823.000000	2823.000000
mean	10258.725115	35.092809	83.658544	6.466171
std	92.085478	9.741443	20.174277	4.225841
min	10100.000000	6.000000	26.880000	1.000000
25%	10180.000000	27.000000	68.860000	3.000000
50%	10262.000000	35.000000	95.700000	6.000000
75%	10333.500000	43.000000	100.000000	9.000000
max	10425.000000	97.000000	100.000000	18.000000

	SALES	QTR_ID	MONTH_ID	YEAR_ID	MSRP
count	2823.000000	2823.000000	2823.000000	2823.000000	2823.000000
mean	3553.889072	2.717676	7.092455	2003.81509	100.715551
std	1841.865106	1.203878	3.656633	0.69967	40.187912
min	482.130000	1.000000	1.000000	2003.00000	33.000000
25%	2203.430000	2.000000	4.000000	2003.00000	68.000000
50%	3184.800000	3.000000	8.000000	2004.00000	99.000000
75%	4508.000000	4.000000	11.000000	2004.00000	124.000000
max	14082.800000	4.000000	12.000000	2005.00000	214.000000

df.dtypes

ORDERNUMBER	int64
QUANTITYORDERED	int64
PRICEEACH	float64
ORDERLINENUMBER	int64
SALES	float64
ORDERDATE	object
STATUS	object
QTR_ID	int64
MONTH_ID	int64
YEAR_ID	int64
PRODUCTLINE	object

```

MSRP                int64
PRODUCTCODE         object
CUSTOMERNAME        object
PHONE               object
ADDRESSLINE1        object
ADDRESSLINE2        object
CITY                object
STATE               object
POSTALCODE          object
COUNTRY             object
TERRITORY           object
CONTACTLASTNAME     object
CONTACTFIRSTNAME    object
DEALSIZE            object
dtype: object

```

```
df.isnull()
```

	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	SALES
\					
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
2818	False	False	False	False	False
2819	False	False	False	False	False
2820	False	False	False	False	False
2821	False	False	False	False	False
2822	False	False	False	False	False

	ORDERDATE	STATUS	QTR_ID	MONTH_ID	YEAR_ID	...	ADDRESSLINE1
\							
0	False	False	False	False	False	...	False
1	False	False	False	False	False	...	False
2	False	False	False	False	False	...	False

3	False	False	False	False	False	...	False
4	False	False	False	False	False	...	False
...	...	...	...	...	...	...	...
2818	False	False	False	False	False	...	False
2819	False	False	False	False	False	...	False
2820	False	False	False	False	False	...	False
2821	False	False	False	False	False	...	False
2822	False	False	False	False	False	...	False

	ADDRESSLINE2	CITY	STATE	POSTALCODE	COUNTRY	TERRITORY	\
0	True	False	False	False	False	True	
1	True	False	True	False	False	False	
2	True	False	True	False	False	False	
3	True	False	False	False	False	True	
4	True	False	False	True	False	True	
...	...	...	...	...	...	...	
2818	True	False	True	False	False	False	
2819	True	False	True	False	False	False	
2820	True	False	True	False	False	False	
2821	True	False	True	False	False	False	
2822	True	False	False	False	False	True	

	CONTACTLASTNAME	CONTACTFIRSTNAME	DEALSIZE
0	False	False	False
1	False	False	False
2	False	False	False
3	False	False	False
4	False	False	False
...	...	...	...
2818	False	False	False
2819	False	False	False
2820	False	False	False
2821	False	False	False
2822	False	False	False

[2823 rows x 25 columns]

df.isnull().sum()

ORDERNUMBER0

QUANTITYORDERED0

PRICEEACH0

ORDERLINENUMBER0

```

SALES          0
ORDERDATE      0
STATUS         0
QTR_ID        0
MONTH_ID       0
YEAR_ID        0
PRODUCTLINE    0
MSRP           0
PRODUCTCODE    0
CUSTOMERNAME    0
PHONE          0
ADDRESSLINE1    0
ADDRESSLINE2    2521
CITY           0
STATE          1486
POSTALCODE      76
COUNTRY         0
TERRITORY       1074
CONTACTLASTNAME 0
CONTACTFIRSTNAME 0
DEALSIZE        0
dtype: int64

```

```

df2 =
df[['ORDERNUMBER', 'QUANTITYORDERED', 'ORDERLINENUMBER', 'QTR_ID', 'MONTH_ID', 'YEAR_ID', 'MSRP']]

```

```
df2
```

	ORDERNUMBER	QUANTITYORDERED	ORDERLINENUMBER	QTR_ID	MONTH_ID
\					
0	10107	30	2	1	2
1	10121	34	5	2	5
2	10134	41	2	3	7
3	10145	45	6	3	8
4	10159	49	14	4	10
...	...	...	...	...	...
2818	10350	20	15	4	12
2819	10373	29	1	1	1
2820	10386	43	4	1	3
2821	10397	34	1	1	3

2822	10414	47	9	2	5
------	-------	----	---	---	---

	YEAR_ID	MSRP
0	2003	95
1	2003	95
2	2003	95
3	2003	95
4	2003	95
...	...	...
2818	2004	54
2819	2005	54
2820	2005	54
2821	2005	54
2822	2005	54

[2823 rows x 7 columns]

df2.dropna()

	ORDERNUMBER	QUANTITYORDERED	ORDERLINENUMBER	QTR_ID	MONTH_ID
\					
0	10107	30	2	1	2
1	10121	34	5	2	5
2	10134	41	2	3	7
3	10145	45	6	3	8
4	10159	49	14	4	10
...	...	...	...	...	...
2818	10350	20	15	4	12
2819	10373	29	1	1	1
2820	10386	43	4	1	3
2821	10397	34	1	1	3
2822	10414	47	9	2	5

	YEAR_ID	MSRP
0	2003	95
1	2003	95
2	2003	95
3	2003	95
4	2003	95

```

...
2818      2004      54
2819      2005      54
2820      2005      54
2821      2005      54
2822      2005      54

```

```
[2823 rows x 7 columns]
```

```
df2.isnull().sum()
```

```

ORDERNUMBER      0
QUANTITYORDERED  0
ORDERLINENUMBER  0
QTR_ID           0
MONTH_ID         0
YEAR_ID          0
MSRP             0

```

```
dtype: int64
```

```
from sklearn.preprocessing import StandardScaler
```

```
scaling = StandardScaler()
```

```
scaled_data = scaling.fit_transform(df2)
```

```
from sklearn.cluster import KMeans
```

```
distortions=[]
```

```
k_range = range(1,12)
```

```
for n in k_range:
```

```
    kmean= KMeans(n_clusters=n, random_state=45, n_init=11)
```

```
    kmean.fit(scaled_data)
```

```
    distortions.append(kmean.inertia_)
```

```
plt.plot(k_range, distortions, 'bx-')
```

```
plt.title("Elbow Method for Clustering")
```

```
plt.xlabel("Number Clusters(k)")
```

```
plt.ylabel("Inertia")
```

```
Text(0, 0.5, 'Inertia')
```

